Marginal boundary energy costs in the Wilson model

Research note

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Abstract

The preceding rough-region note disproved all-small filling with arbitrary frozen boundary links. We now pay for large boundary-forced energy in the *unconditioned* Wilson measure, instead of assuming it can always be removed. On four-dimensional periodic cubes of side $L = 2^m$, $m \geq 2$, site-reflection positivity, the standard event chessboard estimate, and a 64-class link decomposition give an exponential moment bound for the energy of every fixed edited-link set. The same bound controls its conditional minimum in the actual exterior marginal. It also gives a counted tail for source-defined rough-link components, without a boundary compatibility hypothesis. Constants and overlap losses are explicit. These estimates do not produce an all-small compatible filling, an unrestricted lower bound on removable energy, or control of general renormalized or reweighted measures. A sublevel kernel additionally gives partial cooling on a low-minimum class, with a separately paid exceptional label in the raw measure. These are finite Wilson-model inputs, not a continuum construction, mass-gap proof, or novelty claim. The imported chessboard theorem and the locally proved geometric application are distinguished below.

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1 Model and main estimates

Work on $\Lambda_L = (\mathbb{Z}/L\mathbb{Z})^4$, $L = 2^m$, $m \geq 2$. There are $V = L^4$ sites, $4V$ positively oriented links and $6V$ plaquettes. Each link has a variable $U_e \in \mathrm{SU}(2)$, the reverse link has U_e^{-1}, and the reference

measure is product *probability* Haar. Set

$$s(g) = 1 - \tfrac{1}{2}\operatorname{Tr} g = \tfrac{1}{2}\|\mathbf{1} - g\|_{\mathrm{op}}^2, \quad E(U) = \sum_p s(U_p), \quad \mu_H(\,\mathrm{d}U) = Z_H^{-1} e^{-HE(U)} \prod_e \mathrm{d}U_e, \quad H \geq 1.$$

For a fixed link set J, let $n = |J|$ and let $P(J)$ contain *all* plaquettes touching J. Write

$$E_J(U) = \sum_{p \in P(J)} s(U_p), \qquad e_{*,J}(z) = \min_{x \in \mathrm{SU}(2)^J} E_J(x, z), \quad z = U_{J^c}.$$

The minimum exists and is continuous in z. It obeys $0 \leq e_{*,J}(U_{J^c}) \leq E_J(U)$. Define

$$a_H = \tfrac{3}{2} + \log 24 + \tfrac{3}{2}\log(8H), \qquad b_H = \tfrac{1}{2}a_H + \frac{\log 2}{64}. \tag{1}$$

These constants are deliberately loose and differ from the second pass's compatible-kernel constants.

Theorem 1.1 (Unconditioned local energy and boundary cost)**.** *For every fixed J and $0 < \theta < 1$,*

$$\mathbb{E}_{\mu_H} \exp\left(\frac{\theta H}{64} E_J\right) \leq \exp\left[n\left(\theta a_H - \frac{\log(1-\theta)}{64}\right)\right]. \tag{2}$$

The same bound holds with E_J replaced by $e_{*,J}(U_{J^c})$. In particular, for $v \geq 0$,

$$\mu_H(E_J \geq v) \leq \min\{1, e^{-Hv/128 + b_H n}\}, \tag{3}$$

$$\mu_H(e_{*,J}(U_{J^c}) \geq v) \leq \min\{1, e^{-Hv/128 + b_H n}\}. \tag{4}$$

The second probability is under the Wilson exterior marginal*. It is not a statement conditional on an arbitrary fixed exterior.*

For a roughness parameter $p > 0$, call a plaquette rough when $\|\mathbf{1} - U_p\|_{\mathrm{op}} > p/\sqrt{H}$, and write $N_p(J)$ for the number of rough plaquettes in $P(J)$. The letter p here is a parameter, not a plaquette index. All such events are empty if $p/\sqrt{H} \geq 2$.

Corollary 1.2 (Rough counts without boundary compatibility)**.** *For fixed J and a positive integer k,*

$$\mu_H(N_p(J) \geq k) \leq \min\{1, \exp[-kp^2/256 + b_H n]\}. \tag{5}$$

Proof. The event implies $E_J > kp^2/(2H)$. Apply (3). □

These conclusions do not say that the rough plaquettes can be removed while preserving J^c. Their original energy and their boundary-forced minimum are separate quantities.

2 A volume-proportional lower bound on the partition function

Let C_ρ be the geodesic cap in SU(2) of angle at most ρ, with $0 < \rho \leq 1$. The previous note's elementary Haar calculation gives

$$\mathrm{Haar}(C_\rho) = \frac{\rho - \sin\rho\cos\rho}{\pi} \geq \frac{\rho^3}{24}.$$

If every link lies in this cap, each plaquette has operator curvature at most 4ρ, by telescoping its four unitary factors, and hence $E \leq 48V\rho^2$. Take $\rho = (8H)^{-1/2}$. Integrating just over this product cap gives

$$Z_H \geq (\rho^3/24)^{4V} e^{-48HV\rho^2} = e^{-4Va_H}. \tag{6}$$

No independence of plaquette matrices, gauge fixing, or weak-coupling expansion is used. Probability Haar and the nonnegative-action convention are essential to the subsequent numerator bound by one.

3 Site reflection and the chessboard input

Reflect a coordinate x_i to $2a - x_i$ modulo L, where a is an integer. The reflection has two fixed planes, at a and $a + L/2$. An oriented link is transported geometrically; if its orientation reverses, its matrix is inverted. This is a Haar-preserving involution. Tangential links on the fixed planes remain pointwise fixed. Plaquette energy is preserved, because reversal and change of base point invert or conjugate holonomy.

Let z be the links wholly contained in these two planes, and x_+, x_- the remaining links in the two closed half-lattices. No elementary plaquette involves strict interiors on both sides: its extent in the reflected coordinate is zero or one, while the reflecting planes have integer coordinates. Consequently

$$E = E_P(z) + E_+(x_+, z) + E_-(x_-, z), \qquad E_-(x_-, z) = E_+(\vartheta x_-, z).$$

For a bounded real function f supported in the positive closed half,

$$\mathbb{E}_{\mu_H}[f\,\vartheta f] = \frac{1}{Z_H} \int e^{-HE_P(z)} \left[\int e^{-HE_+(x_+,z)} f(x_+, z)\, \mathrm{d}x_+ \right]^2 \mathrm{d}z \geq 0. \tag{7}$$

Reflection symmetry also gives the symmetric bilinear form. This proves site-reflection positivity directly. It does not claim positivity for an arbitrary boundary-conditioned law or for an arbitrary extra interaction.

Imported standard theorem. Tile by closed cubes of side two, $B_t = a + 2t + \{0, 1, 2\}^4$, indexed by $\mathcal{T} = (\mathbb{Z}/(L/2)\mathbb{Z})^4$, with $M = |\mathcal{T}| = V/16$; the origin a can be translated. Neighboring blocks share boundary links. For an event A on the base block, let $\vartheta_t A$ denote its reflected translate to block t: local coordinate v_j becomes v_j for even t_j and $2 - v_j$ for odd t_j, followed by translation by $a + 2t$. The event chessboard inequality is

$$\mu_H \left(\bigcap_{j=1}^{r} \vartheta_{t_j} A_j \right) \leq \prod_{j=1}^{r} \mu_H \left(\bigcap_{t \in \mathcal{T}} \vartheta_t A_j \right)^{1/M}, \qquad t_1, \ldots, t_r \text{ distinct}. \tag{8}$$

We use the event formulation in Biskup–Kotecký [1, §2.2, Theorem 2.4, (2.10)], retaining its power-of-two restriction on $L/2$. Its usual repeated Cauchy–Schwarz proof applies to the closed-half *edge algebras* just described: the needed facts are support under reflection, multiplication, commuting coordinate reflections, and (7). These hold for links at their geometric midpoints, including inversion of reversed orientations. Shared-face variables belong to both closed halves and are integrated as z in (7); they are not independently copied. Thus neither independent blocks nor an identification of outgoing links with spins at their tails is assumed. This is the standard chessboard argument on an edge-decorated lattice, not a new chessboard theorem.

Only the event version (8) is needed below. We derive exponential moments from it explicitly, rather than silently importing a stronger functional version. The geometric extension to these edge algebras and the ensuing analysis are written arguments, not Lean certificates or external expert verification.

4 Reflected stars and the 64 link classes

For a link e, write

$$E_e = E_{\{e\}} = \sum_{q \ni e} s(U_q).$$

A prototype link in a side-two block has direction μ, local starting coordinate $r \in \{0, 1\}$ in direction μ, and starting coordinate one in each of the other three directions. All six incident plaquettes lie in the block. Each of their relative interiors lies inside that block: two coordinates span a unit interval, and the other two equal one. In particular no such plaquette is wholly on a block face.

It follows that the reflected copies of a prototype star have *disjoint plaquette sets*. They may share links on block faces; no probabilistic independence follows or is needed. Their energies obey

$$\sum_{t \in \mathcal{T}} E_{e(t)} \le E. \tag{9}$$

For $u \ge 0$, dissemination of $\{E_e \ge u\}$ forces $E \ge Mu$. Its constrained partition numerator is at most e^{-HMu} since total reference mass is one. Equations (6) and (8) therefore give

$$\mu_H\left(\bigcap_t \{E_{e(t)} \ge u\}\right)^{1/M} \le q_H(u) := \min\{1, \exp[-Hu + 64a_H]\}. \tag{10}$$

The factor 64 here is $4V/M = 4 \cdot 16$, not the number of plaquettes in a star.

Partition all links by their direction and the four parities of their starting site: $4 \cdot 2^4 = 64$ classes. Fix direction μ and parity vector c. Choose the block grid with $a_\mu = c_\mu$ and $a_\nu \equiv c_\nu - 1 \pmod 2$ for $\nu \ne \mu$. Every link of this class is the lower prototype ($r = 0$) in a distinct block of that grid, and all of its incident plaquettes are in that block. When pulling an event back to the base block, odd t_μ replaces this prototype by $r = 1$; both prototypes satisfy (9). Thus (8) allows different pulled-back prototypes.

Lemma 4.1 (Joint star tails within a class)**.** *For a fixed set S in one of these 64 classes and thresholds $u_e \ge 0$,*

$$\mu_H(E_e \ge u_e \text{ for all } e \in S) \le \prod_{e \in S} q_H(u_e). \tag{11}$$

The same bound holds with each E_e replaced by $e_{*,\{e\}}(U_{e^c})$, by event inclusion.

Proof. The events occupy distinct blocks of the common grid. Each reflected prototype has the bound (10). Apply (8). For the final assertion use $e_{*,\{e\}} \le E_e$ pointwise. □

In particular the sharper one-link boundary bound is

$$\mu_H(e_{*,\{e\}} \ge u) \le q_H(u). \tag{12}$$

For the second pass's staple mismatch $\mathcal{D} = \sum_{i<j} \|S_i - S_j\|_{\mathrm{op}}^2$, $e_{*,\{e\}} = \mathcal{D}/(6 + |\sum s_j|) \ge \mathcal{D}/12$, hence

$$\mu_H(\mathcal{D} \ge d) \le \min\{1, e^{-Hd/12 + 64a_H}\}. \tag{13}$$

This pays a marginal statistical cost for large one-link frozen-loop mismatch. It is not an estimate conditional on that mismatch having already been frozen.

5 From joint tails to arbitrary edited sets

Put $K = 64a_H$ and $T_e = HE_e$. Within a class, (11) bounds joint upper tails by products of $q(v) = \min\{1, e^{K-v}\}$. For $0 < \theta < 1$ use

$$e^{\theta T_e} = 1 + \int_0^\infty \theta e^{\theta v} \mathbf{1}_{\{T_e > v\}} \, \mathrm{d}v.$$

Expand a finite product and apply Tonelli and Lemma 4.1 to every subset. The result is

$$\mathbb{E}\exp\left(\theta\sum_{e\in S}T_e\right) \le \left[1+\int_0^\infty \theta e^{\theta v}q(v)\,\mathrm{d}v\right]^{|S|} = \left[\frac{e^{\theta K}}{1-\theta}\right]^{|S|}. \tag{14}$$

The equality follows by splitting the integral at $K \ge 0$. This is an upper-tail comparison, not a claim of independent energies.

Proof of Theorem 1.1. For J, write S_c for its intersections with the 64 classes. Every plaquette in $P(J)$ is counted at least once in $\sum_{e\in J} E_e$, so $E_J \le \sum_{e\in J} E_e$. Apply Hölder with 64 equal exponents and then (14):

$$\begin{aligned}\mathbb{E}e^{\theta H E_J/64} &\le \mathbb{E}\prod_c \exp\left(\frac{\theta}{64}\sum_{e\in S_c}T_e\right)\\ &\le \prod_c\left[\mathbb{E}\exp\left(\theta\sum_{e\in S_c}T_e\right)\right]^{1/64}\\ &\le \exp\left[\frac{n}{64}(\theta K-\log(1-\theta))\right].\end{aligned}$$

Empty classes contribute one; no independence across classes is used. Substitute $K = 64a_H$. The minimum bound follows pointwise from $e_{*,J} \le E_J$. Markov's inequality at $\theta = 1/2$ proves the tails. □

Corollary 5.1 (Explicit overlap loss for retained minima)**.** *Let $J_1,\dots,J_r$ be fixed nonempty link sets, let $I = \bigcup_i J_i$, and let*

$$m_* = \max_{q\in P(I)} |\{i : q \in P(J_i)\}|.$$

Then

$$\mathbb{E}\exp\left[\frac{H}{128m_*}\sum_i e_{*,J_i}(U_{J_i^c})\right] \le e^{b_H|I|}. \tag{15}$$

For disjoint link sets, $m_* \le 4$, not necessarily one.

Proof. Pointwise, $\sum_i e_{*,J_i} \le \sum_i E_{J_i} \le m_* E_I$. Apply (2). A plaquette has four distinct links, so it can touch at most four disjoint J_i. □

This controls a sum of retained energy labels. It does not make energy gains from simultaneous replacements additive.

6 A near-minimum kernel for every boundary

The marginal bound can be paired with a replacement that retains, rather than deletes, the unavoidable minimum. This is a probability-kernel construction, not an efficient sampling algorithm.

Proposition 6.1 (Uniform sublevel replacement)**.** *For every fixed nonempty J there is a Borel gauge-covariant kernel, independent of the original edited links, which freezes J^c, has conditional Haar density at most $(24H^{3/2})^n$, and obeys for every output*

$$E_J(y,z) \le e_{*,J}(z) + 12n/H, \qquad N_p(J;y,z) \le \frac{2He_{*,J}(z)+24n}{p^2}. \tag{16}$$

Its edited-link law depends only on the frozen links of affected plaquettes. It need not remove every rough plaquette or preserve reflection positivity.

Proof. Define

$$B_J(z) = \{y : E_J(y,z) \le e_{*,J}(z) + 12n/H\}, \quad v_J(z) = \mathrm{Haar}_J(B_J(z)), \quad K_z(\,\mathrm{d}y) = \frac{\mathbf{1}_{B_J(z)}(y)}{v_J(z)}\,\mathrm{d}y.$$

To lower-bound v_J, choose any minimizer V for this one fixed z. For $y_e = V_e e^{X_e}$ with $\|X_e\|_{\mathrm{op}} \le \rho = H^{-1/2}$, the second pass's minimizer-cap estimate gives $E_J(y,z) \le e_{*,J}(z) + 12\sum_e \|X_e\|^2$. For clarity, its proof differentiates $E_J(Ve^{tX},z)$ twice: each plaquette contributes at most $(\sum_{e\in\partial q\cap J}\|X_e\|)^2$, and four links per plaquette and six plaquettes per link give $|E_J''(t)| \le 24\sum_e \|X_e\|^2$. The first derivative is zero at an unrestricted minimizer. Taylor's integral remainder supplies the factor 12. Thus $B_J(z)$ contains a product cap of Haar mass at least $(\rho^3/24)^n$, proving the density bound.

Continuity of E_J and $e_{*,J}$ makes the joint sublevel set Borel; parameter integration makes v_J Borel and strictly positive. This proves measurability without choosing a minimizer measurably. A site gauge maps $B_J(z)$ bijectively to $B_J(z^g)$ and preserves edited-link Haar, proving covariance and equality of normalizing volumes. Copying the exterior leaves all unaffected plaquettes unchanged. Finally each rough output plaquette contributes more than $p^2/(2H)$ energy, yielding (16). □

Corollary 6.2 (Partial cooling with a paid exceptional label)**.** *For fixed J and positive integer k, put*

$$A = \{N_p(J) \ge k,\ e_{*,J} \le kp^2/(4H)\}, \qquad \ell_H = \log 24 + \tfrac{3}{2}\log H.$$

If $kp^2 \ge 96n$, the kernel of Proposition 6.1 sends sources in A to at most $3k/4$ rough affected plaquettes, with

$$H(E(x) - E(y)) \ge kp^2/8. \tag{17}$$

If also $kp^2 \ge 16n\ell_H$, then for every nonnegative measurable $\mathcal{Q}$ with $0 < Z_{\mathcal{Q}} = \int e^{-HE}\mathcal{Q} < \infty$,

$$I_{\mathcal{Q}}(A) := \frac{1}{Z_{\mathcal{Q}}}\int_A e^{-HE(x)}(K\mathcal{Q})(x)\,\mathrm{d}x \le e^{-kp^2/16}. \tag{18}$$

The complementary high-minimum label has the raw Wilson *bound*

$$\mu_H(e_{*,J} > kp^2/(4H)) \le \min\{1, e^{-kp^2/512 + b_H n}\}. \tag{19}$$

If $kp^2 \ge 1024 b_H n$, the last expression is at most $e^{-kp^2/1024}$.

Proof. On A, (16) gives $N_p(J;y,z) \le k/2 + 24n/p^2 \le 3k/4$. The source energy is greater than $kp^2/(2H)$ and output energy at most $kp^2/(4H) + 12n/H$, giving $H(E(x)-E(y)) \ge kp^2/4 - 12n \ge kp^2/8$.

For completeness, the bounded-density replacement argument is as follows. Writing $x = (\xi, z)$ and $y = (\eta, z)$, use the pointwise energy drop on A to replace $e^{-HE(\xi,z)}$ by $e^{-kp^2/8}e^{-HE(\eta,z)}$. Bound the edited-output density by $M_H = e^{n\ell_H}$; integrating the original ξ over its probability Haar space costs at most one. Tonelli leaves exactly $Z_{\mathcal{Q}}$ as the remaining integral. Hence $I_{\mathcal{Q}}(A) \le e^{-kp^2/8 + n\ell_H} \le e^{-kp^2/16}$. Apply (4) for (19) and absorb $b_H n$ under the last numerical budget. □

The literal integral in (18) evaluates $\mathcal{Q}$ at the *output*; it is not the $\mathcal{Q}$-weighted probability of A. For $\mathcal{Q} = 1$ it equals the raw probability. The exceptional label in (19) is not paid for an arbitrary $\mathcal{Q}$. Consequently these two inequalities cannot be combined into an unrestricted weighted/RG replacement theorem. Some affected plaquettes may become newly rough: the conclusion concerns their *total count*, not containment in the old rough inventory. Adaptive supports still require the following counting step, and iteration needs gains and density bounds along the actual sequence of outputs.

7 Counting source-dependent rough regions

Connect two lattice links when they share a plaquette. This graph has maximum degree 18. A connected set of n links containing a given root has a deterministic spanning-tree traversal of length $2(n-1)$, and the traversal determines its visited set. Hence there are at most $18^{2(n-1)}$ such sets.

Let $\mathcal{R}_p(U)$ be *all* links incident to at least one rough plaquette. Its connected components are determined by the source configuration. For a root link e_0, let $C_p(e_0)$ be its component, with size zero if $e_0 \notin \mathcal{R}_p$. Put

$$\kappa_H(p) = p^2/1024 - b_H. \tag{20}$$

Theorem 7.1 (Actual rough-component size tail)**.** *For every integer $n \geq 1$,*

$$\mu_H(|C_p(e_0)| \geq n) \leq \min\{1, 18^{2(n-1)} e^{-\kappa_H(p)n}\}. \tag{21}$$

Thus there is an exponentially decreasing size bound when $\kappa_H(p) > 2\log 18$. For $p = (\log H)^P$ and any $P > 1/2$, this condition holds for all sufficiently large H.

Proof. A component of size at least n contains a connected n-link subset J through e_0. Every link in J has a rough witness plaquette in $P(J)$. One plaquette can witness at most four links, so $N_p(J) \geq \lceil n/4 \rceil$. For each *fixed* such J, Corollary 1.2 bounds this event by $\exp[-np^2/1024 + b_H n]$. Sum over the deterministic list of possible J. This treats adaptive selection by a union bound, not by conditioning on the event that a particular J was selected. Finally $b_H = (3/4)\log H + O(1)$ proves the last assertion. □

There is a similar bound for connected retained-minimum labels. If $\lambda = Hu/128 - b_H > 2\log 18$ and $N \geq 1$, then

$$\mu_H(\exists J \ni e_0 \text{ connected},\ |J| \geq N,\ e_{*,J} \geq u|J|) \leq \min\left\{1, \frac{18^{2(N-1)} e^{-\lambda N}}{1 - 18^2 e^{-\lambda}}\right\}. \tag{22}$$

This follows by summing (4) over geometries and sizes. No regular conditional Gibbs formula is applied after choosing J.

8 What this supplies and what it does not

The supplied input. Large boundary-forced energy now has a volume-uniform *marginal* cost in the original finite Wilson model. The one-link holonomy mismatch has the explicit tail (13). The components of the actual rough-link set have a paid source-selection count and the witness density $N_p(J) \geq |J|/4$ needed for (21). Arbitrary overlapping fixed supports can be handled with the declared loss (15). None of these statements assumes the second pass's forest compatibility. Proposition 6.1 also supplies an every-boundary near-minimum kernel. Corollary 6.2 gives a genuine energy gain and rough-count reduction on its specified low-minimum class, with an explicit raw-measure cost for the complementary label.

Not enough removable energy. Outside that low-minimum class, the quantity available to a boundary-preserving edit is at most $E_J - e_{*,J}$. Bounds on the separate marginal tails of E_J and $e_{*,J}$ do not bound their ratio or difference on a selected rough event. The frustrated star still has $E_e = e_{*,\{e\}} = 6$ for every central-link value when its exterior is fixed. Its large-minimum exterior is expensive in (12), but its conditional removal gain is still zero. One cannot replace marginal rarity by conditional repairability. Likewise one isolated $-\mathbf{1}$ link in identity background has total energy 12, regardless of padding the editable region: it cannot pay a volume-proportional replacement entropy in an arbitrarily enlarged set.

A minimum is not a full holonomy label. Take constant commuting links $U_0 = \exp(i\phi\sigma_3/L)$, the other directions identity, with $0 < \phi < \pi$. Every plaquette is flat, so $e_{*,J} = 0$ for every J. Choose J to be the direction-one links starting at $(x_0, 0, 0, 0)$ for all x_0. Its frozen touched graph contains the full direction-zero loop at $x_1 = x_2 = x_3 = 0$, of holonomy $\exp(i\phi\sigma_3) \neq \mathbf{1}$. Thus even zero boundary-forced plaquette energy does not enforce trivial frozen cycles for arbitrary, possibly wrapping, regions. Small-energy estimates cannot silently replace the specified forest compatibility condition.

The raw measure is essential. For $\mathcal{Q} = e^{HE}$, reweighting μ_H gives product Haar. For one plaquette its holonomy is Haar, since its four links are distinct and independent. Therefore $\mu_{H,\mathcal{Q}}(s(U_q) > 1) = 1/2$ for every H, whereas the corresponding raw Wilson tail tends to zero. General positive weights, signed RG expansions, or extra long-range interactions do not inherit this result. Their reflection positivity or a replacement/comparison estimate would have to be proved separately. The bound also does not cover arbitrary fixed boundary conditions, non-dyadic volumes, or all Gibbs states.

Next analytic task. Either strengthen the partial-cooling dichotomy to an all-small, boundary-compatible repair with controlled enlargement and a lower bound on $E_J - e_{*,J}$ per edited link, or build a retained-defect expansion whose weights, normalizers, and interaction changes remain controlled after renormalization. A new kernel must still track all affected plaquettes, its Haar density, output-evaluated weights, and overlap along the actual sequence of edits. Neither task is accomplished by the marginal bounds here. The fixed-physical-box defect test, cutoff comparison, nontrivial continuum fields and reconstruction remain downstream obligations.

9 Verification and literature scope

The accompanying script checks the 64-class embedding, reflected-star plaquette disjointness, the absence of plaquettes crossing both strict halves, SU(2) reflection invariance, the constants and tail integrations, and the obstruction, padding, overlap and flat-holonomy negative controls. It also checks event chessboard inequalities by exact enumeration in a small two-dimensional $\mathbb{Z}_2$ gauge model. That finite-group model is a diagnostic of the bookkeeping, not a numerical evaluation of the four-dimensional SU(2) partition function or a proof of its probabilities. No new analytic result is Lean-formalized; the earlier scalar certificates retain their original scope. This pass is self-audited, not independently agent-reviewed or externally certified.

The only imported analytic inequality is the standard chessboard estimate in [1]. Older gauge-theory literature already discusses volume-uniform suppression of large plaquette fluctuations; see, for context only, [2, §1.1, (4)] and the Mack–Petkova paper cited there. We do not import that

reported bound or its asymptotic constants, and make no novelty claim for the present deliberately weak application. Exact source and access details are recorded in `sources.md`.

References

[1] M. Biskup and R. Kotecký, *Forbidden gap argument for phase transitions proved by means of chessboard estimates*, Commun. Math. Phys. **264** (2006), 631–656, arXiv:math-ph/0505011. Author manuscript, §2.2, Definition 2.2 and Theorem 2.4, (2.10), pp. 4–5.

[2] O. Borisenko, S. Voloshin and M. Faber, *Plaquette representation for 3D lattice gauge models: I. Formulation and perturbation theory*, arXiv:hep-lat/0508003, §1.1, (4), and reference [20]. Context only, not a proof input.